LUCIDMSK CME PROGRAM

MODULE 05 — POST-TEST

Hip MRI & THA: AVN Staging, FAI & Arthroplasty Complications

Hip · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)™

Read each question carefully and select the single best answer. A score of ≥ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 **MRI shows "double line sign" at the anterosuperior femoral head; normal radiograph. Ficat-Arlet stage:**

- **A** Stage I
- **B** Stage IIA
- **C** Stage IIB
- **D** Stage III

2 **Alpha angle measures 68° at the anterosuperior position on radial oblique MRI. This indicates:**

- **A** Normal morphology
- **B** Cam morphology (FAI)
- **C** Pincer morphology
- **D** Hip dysplasia

3 **MARS MRI shows large cystic periarticular fluid collection, thin wall, no solid component around MoM THA. Anderson type:**

- **A** Type 1
- **B** Type 2A (solid)
- **C** Type 2B (cystic)
- **D** Type 4 (destructive)

4 **Post-THA periprosthetic fracture at stem tip with stem subsidence and lucent lines. Vancouver classification:**

- **A** Type AL
- **B** Type B1
- **C** Type B2
- **D** Type C

5 **LCEA measures 22° on standing AP pelvis. Most consistent with:**

A	Normal coverage
B	Pincer FAI (overcoverage)
C	Hip dysplasia
D	Normal variant

6 **MARS MRI shows mixed solid + cystic ALTR around a large-head metal-on-polyethylene THA. Anderson type:**

A	Type 1 — monitor
B	Type 2A — MRI in 6 months
C	Type 3 mixed — early revision
D	Type 4 — emergency surgery

7 **Brooker Grade IV HO is present 9 months post-THA. Next step:**

A	Immediate surgical excision
B	Delayed excision until bone scan quiescence + prophylaxis for contralateral THA
C	Radiation alone
D	Observation only indefinitely

8 **The crescent sign on MRI indicates which ARCO stage?**

A	Stage 1 (pre-radiographic)
B	Stage 2 (sclerosis/cysts)
C	Stage 3 (subchondral fracture)
D	Stage 4 (established OA)

9 **Which THA complication can progress to catastrophic head-neck dissociation?**

A	Particle disease / osteolysis
B	Trunnionosis (taper corrosion)
C	Iliopsoas impingement
D	Heterotopic ossification

10 Vancouver Type C periprosthetic fracture management:

A	Always requires stem revision
B	ORIF without stem revision — fracture distal to stem tip
C	Conservative cast/brace
D	Proximal femoral replacement

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	A	B	C	C	C	C	B	C	B	B

Q1 **MRI shows "double line sign" at the anterosuperior femoral head; normal radiograph. Ficat-Arlet stage:** **Answer: A**

✓ **Stage I**
B Stage IIA
C Stage IIB
D Stage III

EXPLANATION Ficat Stage I = positive MRI (double line sign on T2) with NORMAL radiograph. Stage IIA requires X-ray sclerosis/cysts. Stage IIB = crescent sign. Stage III = femoral head collapse.

Q2 **Alpha angle measures 68° at the anterosuperior position on radial oblique MRI. This indicates:** **Answer: B**

A Normal morphology
✓ **Cam morphology (FAI)**
C Pincer morphology
D Hip dysplasia

EXPLANATION Alpha angle >55° = cam morphology. Normal <55°. 68° is significantly elevated, confirming cam FAI at the anterosuperior head-neck junction.

Q3 **MARS MRI shows large cystic periarticular fluid collection, thin wall, no solid component around MoM THA. Anderson type:** **Answer: C**

A Type 1
B Type 2A (solid)
✓ **Type 2B (cystic)**
D Type 4 (destructive)

EXPLANATION Anderson Type 2B = cystic/fluid-dominant pseudotumor with thin wall, no solid component. Type 2A = solid mass. Type 1 = synovitis without discrete mass. Type 4 = bone/muscle destruction.

Q4 **Post-THA periprosthetic fracture at stem tip with stem subsidence and lucent lines. Vancouver classification:** **Answer: C**

A Type AL
B Type B1
✓ **Type B2**

D Type C

EXPLANATION Vancouver B2 = fracture at/around stem tip with UNSTABLE stem (subsidence + lucent lines = loosening). B1 = same location with stable stem → ORIF. C = distal to stem. AL = lesser trochanter.

Q5 LCEA measures 22° on standing AP pelvis. Most consistent with: **Answer: C**

A Normal coverage

B Pincer FAI (overcoverage)

✓ **Hip dysplasia**

D Normal variant

EXPLANATION LCEA <25° = hip dysplasia (inadequate lateral coverage). Normal = 25–40°. >40° = overcoverage/pincer. Dysplastic hips are at high risk for labral tears and early OA.

Q6 MARS MRI shows mixed solid + cystic ALTR around a large-head metal-on-polyethylene THA. Anderson type: **Answer: C**

A Type 1 — monitor

B Type 2A — MRI in 6 months

✓ **Type 3 mixed — early revision**

D Type 4 — emergency surgery

EXPLANATION Type 3 = mixed solid + cystic ALTR. Large-head MoP hips can develop ALTR via trunnion corrosion. Mixed Type 3 lesions with aggressive features warrant early surgical revision discussion.

Q7 Brooker Grade IV HO is present 9 months post-THA. Next step: **Answer: B**

A Immediate surgical excision

✓ **Delayed excision until bone scan quiescence + prophylaxis for contralateral THA**

C Radiation alone

D Observation only indefinitely

EXPLANATION Grade IV HO excision is delayed until bone maturity (bone scan quiescence, ~12–18 months). Premature excision risks recurrence. Indomethacin or local radiation prophylaxis for contralateral THA.

Q8 The crescent sign on MRI indicates which ARCO stage? **Answer: C**

A Stage 1 (pre-radiographic)

B Stage 2 (sclerosis/cysts)

✓ **Stage 3 (subchondral fracture)**

D Stage 4 (established OA)

EXPLANATION Crescent sign = curvilinear T2 fluid signal (MRI) or lucency (X-ray) = subchondral fracture = ARCO Stage 3. This is the LAST opportunity for joint-preserving surgery before collapse.

Q9 Which THA complication can progress to catastrophic head-neck dissociation? **Answer: B**

A Particle disease / osteolysis

✓ **Trunnionosis (taper corrosion)**

C Iliopsoas impingement

D Heterotopic ossification

EXPLANATION Trunnionosis = progressive corrosion at the femoral head-neck taper junction. If unchecked, it can cause complete head-neck dissociation. Occurs in MoM and large-head MoP bearings.

Q10 Vancouver Type C periprosthetic fracture management: **Answer: B**

A Always requires stem revision

✓ **ORIF without stem revision — fracture distal to stem tip**

C Conservative cast/brace

D Proximal femoral replacement

EXPLANATION Vancouver C = fracture DISTAL to stem tip. The stem is uninvolved. Standard ORIF (plate ± cerclage) without stem revision — even if stem shows some loosening. Key distinction from B2.